On the Impact of Graph Neural Networks in Recommender Systems

A Topological Perspective

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ABSTRACT

In recommender systems, user–item interactions can be modeled as a bipartite graph, where user and item nodes are connected by undirected edges. This graph-based view has motivated the rapid adoption of graph neural networks (GNNs), which often outperform collaborative filtering (CF) methods such as latent factor models, deep neural networks, and generative strategies. Yet, despite their empirical success, the reasons why GNNs offer systematic advantages over other CF approaches remain only partially understood.

This monograph advances a topology-centered perspective on GNN-based recommendation. We argue that a comprehensive understanding of these models' performance should consider the structural properties of user–item graphs and their interaction with GNN architectural design. To support this view, we introduce a formal taxonomy that distills common modeling patterns across eleven representative GNN-based recommendation approaches and consolidates them into a unified conceptual pipeline.

We further formalize thirteen classical and topological characteristics of recommendation datasets and reinterpret them

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through the lens of graph machine learning. Using these definitions, we analyze the considered GNN-based recommender architectures to assess how and to what extent they encode such properties. Building on this analysis, we derive an explanatory framework that links measurable dataset characteristics to model behavior and performance.

Taken together, this monograph re-frames GNN-based recommendation through its topological underpinnings and outlines open theoretical, data-centric, and evaluation challenges for the next generation of topology-aware recommender systems.